

Aaron Danen

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Education

University of California, Davis – B.S. in Computer Science and Engineering, 3.97 GPA Expected June 2027

Coursework: Data Structures and Algorithms, Computer Architecture, Operating Systems, Programming Languages

Honors: Dean's List, Tau Beta Pi Honor Society

Skills

Languages: C, C++, Python, LLVM IR, x86 Assembly

Tools: git, clang/gcc, cppcheck, CMake, Make, Linux, Cygwin

Libraries: LLVM, Pandas, Matplotlib, Numpy, SDL3

Experience

Compilers Intern, Hewlett Packard Enterprise June 2026 - Present

- Modified preprocessor, parser, and dataflow analysis of open source static analysis tool Cppcheck to support Nonstop C/C++ extensions such as optional function arguments, multiple size pointers, and compiler intrinsics
- Discovered and reported real memory safety bugs in Nonstop software using modified Cppcheck analyzer
- Iterated with QA engineers to develop CLI features, and workflows optimal for analyzing Nonstop software
- Resolved 6 upstream bugs by producing minimal examples, finding the root cause, and implementing solutions

Data Engineering Analyst, UC Santa Cruz, Dept. of Astrophysics Aug 2022 - Sept 2023, Oct 2024 - Aug 2025

- Automated data retrieval and preprocessing pipelines for VERITAS and NICER telescope data using Python and Pandas, reducing manual labor by 10+ hours per week for researchers seeking up-to-date measurements
- Automated analysis and aggregation of data from 4 observatories, contributing to a \$90,000 grant proposal
- Surveyed and met users visualization needs by enriching the data through statistical methods and using Matplotlib
- Leveraged existing NASA CLI tools into a Python interface with shell scripts and string processing algorithms

Open Source Contributions

BJET_MCMC, Open Source Statistical Analysis Software github.com/ohervet/Bjet_MCMC

- Improved runtime by 20% by running tests on a OpenHPC cluster to optimize Monte Carlo Markov Chain parameters
- Authored performance and optimization documentation page to save users computation time
- Contributed a feature that fits an exponential decay function to the Markov Chain convergence and reports to users

LLVM, Open Source Compiler Framework github.com/llvm/llvm-project/pull/150348

- Strengthened users experiences with the clang-repl tool by collaborating with other contributors to identify weaknesses, implementing a %help command in C++ list commands, and clarified error messages
- Achieved a fast merge into the main branch by studying contribution guidelines and following coding standards

Projects

CHIP-8 Interpreter github.com/aadanen/chip8

- Developed an interpreter for a classic retro programming language that supports all standard ROMs and automatically adjusts its behavior to resemble that of any major CHIP-8 interpreter from the previous half-century
- Streamlined the build process to just a few commands by using CMake and minimizing dependencies
- Crafted a retro audiovisual experience with graphics, audio, and user input using SDL3

Kalos, A Compiler for LLVM's Kaleidoscope Language github.com/aadanen/llvm-kaleidoscope

- Studied lexical analysis and parsing theory before building a compiler by researching LLVM's IR and APIs
- Enabled users to incorporate Kaleidoscope into other programs by compiling to standard object files

Activities

UC Davis Competitive Programming Club President

California All-State High School Honor Jazz Band